AZURE SERVICES QUICK REFERENCE

AZ-900 + AZ-104 | Concept · Why · Workflow · Best For · Key Metric

Coverage Legend: **[Both]** appears in both courses | **[AZ-900]** AZ-900 only | **[AZ-104]** AZ-104 only

◆ COMPUTE SERVICES

Azure Virtual Machine (VM) [Both]

Concept: An on-demand, scalable IaaS compute resource running Windows or Linux in Azure datacenters. Equivalent: Node (in K8s terms — a raw compute unit you manage yourself)

Primary Why & What: Run any workload that requires full OS control — web servers, databases, custom runtimes — without managing physical hardware. You choose the size, OS, disk, and network.

Workflow: Input: Choose OS image + VM size + region → Process: Azure provisions vCPU, RAM, OS disk, NIC in a VNet → Output: A running machine accessible via RDP (Windows) or SSH (Linux)

Best For: Hosting a legacy enterprise application (e.g., a .NET app on IIS) that cannot be containerised and requires dedicated Windows Server resources

⚡ **Key Metric: Billed per second based on VM size (vCPU + RAM tier). Stops billing when Deallocated — NOT when just Stopped (OS-level)**

Azure Virtual Machine Scale Sets (VMSS) [Both]

Concept: A group of identical, load-balanced VMs that auto-scale in and out based on demand. Equivalent: ReplicaSet / HPA in Kubernetes

Primary Why & What: Eliminates manual scaling. When traffic spikes, Azure adds VMs automatically; when traffic drops, it removes them — keeping costs proportional to load.

Workflow: Input: Base VM image + scaling rules (CPU %, schedule, metrics) → Process: Azure monitors metrics and provisions or removes identical VM instances → Output: Always-appropriate pool of VMs behind a load balancer

Best For: A retail website during a flash sale — VMSS scales from 2 VMs to 20 in minutes as traffic surges, then scales back down automatically

⚡ **Key Metric: Billed per VM instance active at any time. Flexible orchestration mode allows mixed instance types**

Azure Web Apps (App Service) [Both]

Concept: A fully managed PaaS platform to host web applications, REST APIs, and mobile backends. Equivalent: No direct K8s equivalent — closest is an Ingress + Deployment managed for you

Primary Why & What: Run web apps without managing the OS, patching, or runtime environment. Azure handles scaling, certificates, and deployment pipelines. You only manage your code.

Workflow: Input: Code (Git, ZIP, container image) + App Service Plan (size) → Process: Azure builds runtime env, deploys code, maps domain → Output: Live web app on *.azurewebsites.net or custom domain

Best For: Deploying a Node.js or .NET API where the team has zero interest in managing servers — just push code to a Git repo and it's live

⚡ Key Metric: Billed per App Service Plan tier (Basic, Standard, Premium). Multiple apps can share one plan at no extra cost

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Azure Functions [AZ-900]

Concept: A serverless, event-driven compute service that runs individual functions triggered by events. Equivalent: No direct Docker equivalent — closest conceptual match is a single-purpose sidecar container triggered by events

Primary Why & What: Run small pieces of code on demand without provisioning any server. You only pay when the function actually executes. Perfect for glue code, event handlers, and automation tasks.

Workflow: `Input: Trigger event (HTTP request, queue message, blob upload, timer) → Process: Azure spins up an isolated runtime, executes your function code → Output: Response or side-effect (e.g., writes to DB), then runtime tears down`

Best For: Processing an image every time it is uploaded to Azure Blob Storage — the function triggers automatically, resizes the image, and saves the thumbnail

⚡ Key Metric: Billed per execution count + GB-second of memory consumed. First 1M executions/month are free on Consumption plan

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Azure Container Registry (ACR) [AZ-104]

Concept: A private, managed Docker registry hosted on Azure for storing and managing container images. Equivalent: Docker Hub (private) — a centralised image store

Primary Why & What: Store your organisation's container images securely in Azure, close to your AKS clusters and Azure Container Instances, avoiding public registry exposure.

Workflow: `Input: Local Docker image built from Dockerfile → Process: docker push to ACR / az acr build → Output: Versioned container image stored in private registry, ready to pull`

Best For: A company whose development team builds a PHP app image locally and needs to push it securely to a private registry that ACI or AKS can pull from without public internet exposure

⚡ Key Metric: Billed per registry tier (Basic/Standard/Premium) + GB of image storage used

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Azure Container Instances (ACI) [Both]

Concept: Run containers directly on Azure without managing any underlying VM or orchestrator. Equivalent: docker run — the simplest way to run a container in the cloud

Primary Why & What: Fastest path from a container image to a running workload in Azure. No cluster to manage, no scheduler to configure. Ideal for short-lived, isolated tasks.

Workflow: `Input: Container image (from ACR or Docker Hub) + resource limits → Process: Azure provisions an isolated container environment → Output: Running container accessible via public IP or private VNet`

Best For: Running a one-off data processing job that reads from a storage account, transforms data, writes results, and exits — no always-on server needed

⚡ Key Metric: Billed per second of vCPU + GB of memory while the container is running. Billing stops the moment the container exits

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Azure Container Apps [AZ-104]

Concept: A serverless, fully managed platform for running microservices and containerised applications with built-in scaling (including scale-to-zero). Equivalent: Knative on top of Kubernetes — managed serverless containers

Primary Why & What: Run containerised microservices without managing Kubernetes clusters. Supports event-driven scaling, Dapr integration, and HTTP-triggered auto-scaling out of the box.

Workflow: `Input: Container image + environment config + scaling rules → Process: Azure manages the underlying K8s infra, routes traffic, scales instances → Output: Fully managed containerised app reachable via HTTPS`

Best For: A team deploying a set of microservices that need independent scaling — some services scale to zero overnight, others scale to 50 instances at peak

⚡ **Key Metric: Billed per vCPU-second + GB-second consumed, plus per request count. Scales to zero = zero cost when idle**

Azure Virtual Desktop [AZ-900]

Concept: A cloud-hosted desktop and application virtualisation service — delivers full Windows desktops to users via browser or client. Equivalent: No container/Docker equivalent — it is VDI (Virtual Desktop Infrastructure) as a service

Primary Why & What: Give remote workers a fully managed Windows 10/11 desktop without shipping physical hardware. Centralise security, patching, and management in Azure.

Workflow: `Input: Host pool definition + workspace + Windows image → Process: Azure provisions Windows VMs, sessions are brokered to end users → Output: Full Windows desktop streamed to any device via RDP client or browser`

Best For: A company onboarding 200 remote contractors who need a secure Windows environment — no laptops shipped, desktops are ready in Azure within hours

⚡ **Key Metric: Billed per underlying VM compute hours + storage. Multi-session Windows allows multiple users per VM reducing per-user cost**

Azure Batch Service [AZ-900]

Concept: A managed service to run large-scale parallel and high-performance computing (HPC) batch jobs across a pool of VMs. Equivalent: A managed job scheduler — no direct Docker equivalent

Primary Why & What: Process massive, parallelisable workloads (video encoding, simulations, data transformation) without manually managing a fleet of VMs. Azure Batch handles provisioning, scheduling, retries, and cleanup.

Workflow: `Input: Job definition + task list + input data + pool spec → Process: Azure provisions pool VMs, distributes tasks, monitors completion → Output: Results written to storage; pool can be auto-deleted when done`

Best For: A media platform that needs to transcode 10,000 uploaded videos into 4 formats each night — Batch spins up 200 VMs, processes all jobs in parallel, and tears them down by morning

⚡ **Key Metric: Billed per VM compute time in the pool. No charge for the Batch service itself — only underlying VM + storage costs**

◆ NETWORKING SERVICES

Azure Virtual Network (VNet) [Both]

Concept: A logically isolated private network in Azure where you control the IP address space, subnets, routing, and gateways. Equivalent: Docker network (bridge/overlay) — your private layer-2/3 environment

Primary Why & What: All Azure resources that need to communicate privately must live inside a VNet. Without it, VMs have no private network to talk to each other securely.

```
Workflow:  Input: Define address space (CIDR, e.g. 10.0.0.0/16) + subnets → Process: Azure creates isolated network fabric → Output: Resources deployed to subnets get private IPs and can communicate securely
```

Best For: Isolating a web tier (public subnet) from a database tier (private subnet) within a single application — DB VMs are unreachable from the internet by default

⚡ **Key Metric: VNet itself is free. Billed for outbound data transfer, VNet peering, and gateways**

Network Security Group (NSG) [Both]

Concept: A virtual firewall that filters inbound and outbound traffic to/from Azure resources using priority-ordered rules. Equivalent: iptables / Kubernetes NetworkPolicy

Primary Why & What: Control exactly which traffic is allowed into and out of your VMs and subnets. Without NSGs, resources are exposed to all traffic from within the VNet and potentially from the internet.

```
Workflow:  Input: Rule definition (priority, source IP, destination port, protocol, allow/deny) → Process: Azure evaluates rules in priority order for each packet → Output: Traffic is permitted or blocked at the network layer
```

Best For: Blocking all internet traffic to a database VM except port 3306 from the web tier subnet — enforcing least-privilege network access

⚡ **Key Metric: Billed per NSG + per 10,000 rules per month. Associating NSG to subnet vs NIC has different scope — NSG on subnet applies to ALL resources in that subnet**

Azure Bastion [Both]

Concept: A fully managed PaaS service that provides secure RDP/SSH connectivity to VMs directly through the Azure Portal over TLS, without exposing public IPs. Equivalent: A jump-box / bastion host — managed by Azure

Primary Why & What: Eliminates the need to expose VMs to the internet via public IPs or open port 3389/22 on NSGs. All connections go through the Azure portal browser session — no additional client software needed.

```
Workflow:  Input: Deploy Azure Bastion to a subnet (AzureBastionSubnet) in the VNet → Process: Azure proxies RDP/SSH traffic through its managed service → Output: Browser-based terminal/desktop session to private-IP VMs
```

Best For: A production environment where VMs have no public IPs — admins connect securely via Azure Portal without opening any firewall ports

⚡ **Key Metric: Billed per Bastion host per hour + outbound data transfer. Basic vs Standard SKU determines max sessions and native client support**

Azure Load Balancer [Both]

Concept: A Layer 4 (TCP/UDP) load balancer that distributes inbound traffic across a pool of backend VMs. Equivalent: Kubernetes Service (type: LoadBalancer) at the network layer

Primary Why & What: Distribute traffic evenly across VMs to eliminate single points of failure and improve throughput. Standard SKU also enables outbound SNAT for VMs without public IPs.

Workflow: `Input: Frontend IP (public or private) + backend pool (VMs/VMSS) + health probe + load balancing rules → Process: Azure routes each connection to the healthiest backend instance → Output: High-availability endpoint for your VM pool`

Best For: Distributing HTTP traffic across 3 web server VMs — if one VM fails its health probe, the load balancer automatically stops sending traffic to it

⚡ **Key Metric: Billed per rule per hour + data processed. Standard SKU required for Availability Zones and VMSS integration**

Azure Traffic Manager [AZ-900]

Concept: A DNS-based global traffic load balancer that routes users to the best endpoint (by performance, geography, priority, or weighted distribution). Equivalent: No K8s equivalent — operates at DNS layer, not network layer

Primary Why & What: Route users to the nearest or healthiest deployment across multiple Azure regions (or even on-premises endpoints). Unlike Load Balancer, it works at the DNS level — not the packet level.

Workflow: `Input: Profile with routing method + endpoints (Azure services, external URLs) + health checks → Process: DNS query comes in, Traffic Manager returns the IP of the best endpoint based on routing policy → Output: Client connects directly to the selected endpoint`

Best For: A globally deployed web app with instances in East US and West Europe — Traffic Manager sends US users to East US and European users to West Europe for minimum latency

⚡ **Key Metric: Billed per DNS query + per endpoint health check per month**

Azure Front Door [AZ-900]

Concept: A global, scalable CDN and Layer 7 (HTTP) load balancer with WAF, SSL offloading, and intelligent routing. Equivalent: No K8s equivalent — think NGINX Ingress + CDN + WAF combined globally

Primary Why & What: Deliver web content with low latency globally by caching at edge PoPs worldwide, while routing traffic intelligently and protecting against web application attacks.

Workflow: `Input: Backend origins (Web Apps, VMs, storage) + routing rules + WAF policies → Process: Azure Front Door edge PoP nearest to user handles request, caches or proxies to origin → Output: Fast, secure content delivery to global users`

Best For: A SaaS platform with users across Asia, Europe, and Americas — Front Door caches static assets at edge locations globally, reducing origin load and cutting latency from 400ms to 30ms for distant users

⚡ **Key Metric: Billed per GB outbound data transferred + number of routing rules + WAF requests**

Azure Firewall [AZ-900]

Concept: A fully managed, stateful network security service that inspects and filters traffic flowing in and out of Azure Virtual Networks. Equivalent: No direct K8s equivalent — think of it as a managed enterprise-grade firewall appliance in the cloud

Primary Why & What: NSGs are rule-based and operate per resource. Azure Firewall provides centralised, organisation-wide policy enforcement with FQDN filtering, threat intelligence, and east-west traffic inspection.

Workflow: `Input: Deploy Azure Firewall to a dedicated subnet (AzureFirewallSubnet) + define DNAT/network/application rules → Process: All traffic routed through firewall via UDRs; firewall inspects and allows or drops → Output: Policy-compliant ingress/egress for entire VNet`

Best For: An enterprise that wants all outbound internet traffic from their VNet to be inspected and only allow specific FQDNs (e.g., *.microsoft.com) — blocking all other destinations

Key Metric: Billed per deployment hour + per GB of data processed through the firewall

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Azure DDoS Protection [AZ-900]

Concept: A service that detects and mitigates Distributed Denial-of-Service attacks targeting public IP resources in Azure. Equivalent: No K8s equivalent — operates at the Azure infrastructure/network edge

Primary Why & What: Protect public-facing applications from volumetric, protocol, and application-layer DDoS attacks that could make services unavailable by flooding them with malicious traffic.

Workflow: `Input: Enable DDoS protection on a VNet (Network Protection) or individual Public IP → Process: Azure monitors traffic patterns in real-time; on attack detection, traffic is scrubbed at the edge → Output: Legitimate traffic reaches your app; attack traffic is dropped`

Best For: A financial services company hosting a public API — DDoS protection automatically deflects a 500 Gbps UDP flood attack without any manual intervention

Key Metric: Network Protection billed per protected VNet per month (flat fee) + overage per GB scrubbed. IP Protection billed per protected public IP per month

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Azure VPN Gateway [AZ-900]

Concept: A service that creates encrypted tunnels between Azure VNets and on-premises networks (or other VNets) over the public internet using IPSec/IKE protocols. Equivalent: No K8s equivalent — WireGuard/IPSec tunnel at infrastructure level

Primary Why & What: Securely extend an on-premises network into Azure — employees at the office can access Azure VMs using private IP addresses as if they were on the same local network.

Workflow: `Input: VPN Gateway deployed in GatewaySubnet + local network gateway (on-prem IP + address space) + shared key → Process: Encrypted IPSec tunnel established over internet → Output: Private connectivity between on-premises and Azure VNet`

Best For: A company migrating workloads to Azure that needs their on-premises ERP system to communicate securely with new Azure VMs using private IPs during the migration period

Key Metric: Billed per gateway per hour (SKU-based) + outbound data transfer

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Azure ExpressRoute [AZ-900]

Concept: A private, dedicated, high-bandwidth connectivity service between on-premises networks and Azure — does NOT go over the public internet. Equivalent: A dedicated physical fibre line to Azure's datacentre

Primary Why & What: For enterprises requiring guaranteed bandwidth, consistent low latency, and private (non-internet) connectivity — typically for compliance, large data transfers, or mission-critical workloads.

Workflow: `Input: Work with connectivity provider + create ExpressRoute circuit + link to VNet gateway → Process: Traffic flows via provider's MPLS network`

directly to Azure edge → Output: Private, high-speed, SLA-backed connection to Azure

Best For: A bank that must transfer multi-TB nightly database backups to Azure without the data ever touching the public internet — required by their regulatory compliance policy

⚡ **Key Metric: Billed per circuit bandwidth tier (50 Mbps to 100 Gbps) per month + data transfer (metered circuits)**

Azure Private DNS [Both]

Concept: A managed DNS service for resolving private (non-internet) domain names within Azure Virtual Networks. Equivalent: CoreDNS in Kubernetes — internal name resolution for private resources

Primary Why & What: Let Azure resources resolve each other by custom domain names (e.g., db.internal.contoso.com → 10.0.1.5) without setting up and managing your own DNS server.

Workflow: Input: Create Private DNS Zone + link to VNet(s) + add DNS A records → Process: Azure DNS resolves queries from linked VNets using the private zone → Output: VMs resolve private FQDNs to internal IP addresses

Best For: An AKS cluster needing to resolve a private SQL server by name (sql.prod.internal) rather than by IP — making config files environment-agnostic

⚡ **Key Metric: Billed per DNS zone per month + per million queries**

Azure Network Watcher [AZ-104]

Concept: A suite of network diagnostic and monitoring tools for Azure virtual networks. Equivalent: tcpdump + traceroute + Wireshark — but as managed cloud tools

Primary Why & What: Diagnose and troubleshoot network connectivity problems between VMs, across VNets, or to the internet — without needing to SSH into machines or install agents.

Workflow: Input: Select source + destination + tool (Connection Troubleshoot / IP Flow Verify / Next Hop / NSG Flow Logs) → Process: Azure runs the diagnostic against the live network state → Output: Pass/Fail verdict + hop path + NSG rule responsible for blocking

Best For: Diagnosing why a VM in Subnet A cannot reach a VM in Subnet B — IP Flow Verify identifies the exact NSG rule blocking the traffic in under 10 seconds

⚡ **Key Metric: Most tools are free. NSG Flow Logs billed per GB of logs stored**

◆ STORAGE SERVICES

Azure Storage Account [Both]

Concept: A top-level Azure resource that groups together four storage services: Blob, File, Queue, and Table — all under one namespace and one set of access credentials. Equivalent: A storage namespace — like a PersistentVolume claim's backing store in K8s

Primary Why & What: All cloud applications need durable, scalable, cheap object/file/queue storage. Azure Storage Account is the foundation for virtually all data storage outside of managed databases.

Workflow: Input: Define account name + redundancy type (LRS/ZRS/GRS) + access tier (Hot/Cool) → Process: Azure provisions globally unique namespace with 4 sub-services → Output: Endpoints available (blob.core.windows.net, file.core.windows.net, etc.)

Best For: Storing all static assets (images, videos, documents) for a web application in Blob storage — globally accessible, infinitely scalable, zero server management

⚡ Key Metric: Billed per GB stored + per operation (read/write/list) + per egress GB. Access tier (Hot vs Cool vs Archive) significantly affects both storage and retrieval costs

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Azure Blob Storage [Both]

Concept: The object storage service within a Storage Account — store unstructured data (images, videos, logs, backups, any file format) as 'blobs' in 'containers'. Equivalent: Docker volume (for files) — but cloud-scale object store

Primary Why & What: Infinitely scalable, durable object storage with no filesystem overhead. The go-to solution for storing any binary or text data that doesn't need a relational structure.

Workflow: `Input: Create container → Upload blob (Block/Append/Page blob type) → Set access level (private/public) → Process: Azure stores with 3+ replicas → Output: Blob accessible via URL or SDK`

Best For: Storing thousands of user-uploaded profile pictures for a mobile app — each image gets a unique URL, auto-replicated for durability, and served at scale without any server configuration

⚡ Key Metric: Billed per GB stored (varies by tier: Hot = high cost, Archive = lowest cost) + per-operation charges + retrieval fees for Cool/Archive tiers

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Azure File Shares (Azure Files) [Both]

Concept: A fully managed, cloud-hosted SMB/NFS file share — like a network drive in the cloud that can be mounted on Windows, Linux, or macOS. Equivalent: PersistentVolume with ReadWriteMany access mode in Kubernetes

Primary Why & What: Lift-and-shift legacy applications that depend on shared network drives (mapped drives, UNC paths). Multiple VMs can mount the same file share simultaneously — no NAS server to manage.

Workflow: `Input: Create file share in storage account → Generate mount command (SMB 3.0) → Process: VMs mount share using storage account key → Output: Shared network drive accessible from multiple machines simultaneously`

Best For: Multiple application servers all writing log files to a shared location — Azure File Share replaces the physical NAS, is accessible from all VMs, and persists when VMs are deallocated

⚡ Key Metric: Billed per GB used per month (Transaction Optimised / Hot / Cool tiers) + per operation. Premium file shares billed per provisioned GB

Azure File Sync [AZ-900]

Concept: A service that synchronises on-premises Windows file servers with Azure File Shares, enabling cloud tiering (infrequently used files moved to cloud, recalled on access). Equivalent: rsync + inotify — but managed and bidirectional

Primary Why & What: Organisations with on-premises file servers can centralise backup and management in Azure while keeping local performance for frequently accessed files — without rearchitecting existing workflows.

Workflow: `Input: Register Windows Server + define sync group + configure cloud endpoint (Azure File Share) → Process: Azure File Sync agent syncs files bidirectionally; cloud tiering moves cold files to Azure → Output: On-premises server shows all files (stubs for cold ones), actual data in Azure`

Best For: A branch office with a Windows file server that needs to back up everything to Azure while ensuring frequently used project files stay fast and locally accessible

Key Metric: Billed per server registered + per GB synced + cloud tiering policy (if enabled)

Azure Queue Storage [AZ-900]

Concept: A simple, highly available message queue service for storing large numbers of messages and decoupling application components. Equivalent: A lightweight message bus — similar concept to a Kubernetes liveness probe trigger but for async processing

Primary Why & What: Decouple producers from consumers in distributed applications. The producer writes a message to the queue and moves on — the consumer reads and processes at its own pace.

```
Workflow:  Input: Producer writes message to queue → Process: Message persists (up to 7 days) until consumed; consumer reads message, processes, then deletes it → Output: Reliable async communication between app components
```

Best For: An order processing system where a web app posts new orders to a queue, and a separate worker service picks them up and processes payments asynchronously — web app never waits for payment completion

Key Metric: Billed per 10,000 operations (puts, gets, deletes). Storage cost per GB of queue data

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Azure Table Storage [AZ-900]

Concept: A NoSQL key-value/columnar datastore for structured, non-relational data — stores data as entities (rows) with flexible columns under a partition key + row key model. Equivalent: etcd (key-value store) in spirit, but tabular

Primary Why & What: Store structured datasets that don't need joins, foreign keys, or stored procedures — at massive scale and very low cost compared to SQL databases.

```
Workflow:  Input: Define table + insert entities (PartitionKey + RowKey + custom properties) → Process: Azure distributes data across partitions for scale → Output: Query entities by PartitionKey/RowKey via REST API or SDK
```

Best For: Storing user session data or device telemetry for an IoT application — millions of records, simple key-based lookups, no complex queries, dirt cheap per GB

Key Metric: Billed per GB stored + per 10,000 transactions. Significantly cheaper than Azure SQL for simple read/write workloads

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Azure Data Box [AZ-900]

Concept: A physical appliance shipped to your premises by Microsoft for bulk offline data transfer to/from Azure — when internet-based transfer would take too long or be too costly. Equivalent: No Docker/K8s equivalent — it is a physical USB-like device at datacentre scale

Primary Why & What: Transferring petabytes of data over the internet takes weeks and costs a fortune in egress fees. Data Box lets you copy data to the device, ship it back to Microsoft, and they upload it directly to Azure Storage.

```
Workflow:  Input: Order Data Box device via Azure Portal → Receive device + connect to on-premises network → Copy data → Ship device back to Microsoft → Process: Microsoft uploads data to your Azure Storage Account → Output: Petabytes of data in Azure without internet transfer
```

Best For: A hospital migrating 500 TB of medical imaging archives to Azure Blob Storage — internet transfer would take 4+ months; Data Box does it in 10 days

⚡ **Key Metric: Billed per device rental + shipping. No data egress charges. Sizes: Data Box (100 TB), Data Box Heavy (1 PB)**

◆ DATABASE SERVICES [AZ-900 Coverage]

Azure SQL Database [AZ-900]

Concept: A fully managed, cloud-native relational database service based on Microsoft SQL Server — PaaS, no infrastructure to manage. Equivalent: No K8s equivalent — managed SQL as a service

Primary Why & What: Run production relational databases without patching OS, SQL Server binaries, or managing high availability. Azure handles backups, failover, and scaling automatically.

Workflow: `Input: Define server + database + pricing tier (DTU or vCore) → Process: Azure provisions managed SQL Server instance → Output: SQL endpoint accessible via connection string — query with T-SQL immediately`

Best For: A .NET web application needing a production SQL Server database without a DBA team — Azure SQL handles HA, backups, and patching with zero manual intervention

⚡ **Key Metric: Billed per DTU (Database Transaction Unit) or per vCore + storage per GB. Serverless tier scales compute to zero when idle**

Azure Database for MySQL [AZ-900]

Concept: A fully managed MySQL database service on Azure — same engine, no infrastructure management. Equivalent: No K8s equivalent — managed open-source MySQL as PaaS

Primary Why & What: Run MySQL workloads in the cloud with enterprise features (HA, backups, read replicas) without managing MySQL server installation, updates, or failover infrastructure.

Workflow: `Input: Choose compute tier + storage size + MySQL version → Process: Azure provisions managed MySQL server with HA option → Output: Standard MySQL endpoint — connect with any MySQL client or ORM`

Best For: A PHP/Laravel web application already using MySQL on-premises — lift and shift the database to Azure Database for MySQL with minimal code changes

⚡ **Key Metric: Billed per vCore + GB of storage + backup storage. Flexible Server pricing model is current recommended tier**

Azure Cosmos DB [AZ-900]

Concept: A globally distributed, multi-model NoSQL database service with single-digit millisecond response times and automatic multi-region replication. Equivalent: No K8s equivalent — think Redis + MongoDB + Cassandra combined, globally distributed

Primary Why & What: For applications requiring planet-scale throughput, global distribution, and flexible data models (document, key-value, graph, column-family). Guarantees latency SLAs no SQL database can match.

Workflow: `Input: Create account + choose API (SQL/Core, MongoDB, Cassandra, Gremlin, Table) + configure regions + set throughput (RU/s) → Process: Data auto-replicated to chosen regions with configurable consistency → Output: Sub-10ms reads globally`

Best For: A real-time gaming leaderboard serving 10M concurrent users worldwide — Cosmos DB replicates data to 5 regions, guarantees <10ms reads globally, and scales throughput on demand

⚡ **Key Metric: Billed per Request Unit/second (RU/s) provisioned + per GB stored + per region replicated to. Serverless option bills per RU consumed**

Azure Synapse Analytics [AZ-900]

Concept: An integrated analytics service combining SQL data warehousing, Apache Spark big data processing, and data integration pipelines. Equivalent: No K8s equivalent — enterprise data warehouse + ETL engine as a service

Primary Why & What: Unify data ingestion, transformation, and analysis in one service. Run complex analytical SQL queries over petabytes of data without managing the underlying distributed compute infrastructure.

```
Workflow:  Input: Ingest data from Data Lake / databases via pipelines →
Process: Dedicated SQL Pool (MPP warehouse) or Spark Pool processes queries →
Output: Analytical results queryable via Power BI or other reporting tools
```

Best For: A retail company analysing 3 years of transaction data (50 TB) to find seasonal buying patterns — Synapse runs MPP SQL queries across the full dataset in minutes, not hours

⚡ **Key Metric: Billed per DWU (Data Warehouse Unit) for Dedicated Pool + per vCore-hour for Spark Pools + pipeline activity runs**

Azure Databricks [AZ-900]

Concept: A managed Apache Spark analytics platform built for large-scale data engineering, machine learning, and collaborative notebooks. Equivalent: No K8s equivalent — Apache Spark as a managed cloud service

Primary Why & What: Process and transform massive datasets using distributed Spark jobs without managing Spark cluster setup, Hadoop dependencies, or version compatibility — created by the creators of Apache Spark.

```
Workflow:  Input: Data in Azure Data Lake / Blob Storage → Process: Spark
cluster processes data using Python/Scala/SQL notebooks → Output: Cleaned,
transformed data written back to lake or warehouse
```

Best For: A data engineering team that needs to run daily ETL jobs transforming 10 TB of raw JSON event logs into a structured Parquet format for analytics — Databricks Spark jobs do this in minutes

⚡ **Key Metric: Billed per Databricks Unit (DBU) per hour — rate depends on cluster tier (Standard, Premium) and VM size**

◆ IDENTITY AND ACCESS MANAGEMENT

Microsoft Entra ID (formerly Azure AD) [Both]

Concept: Azure's cloud-based identity and access management service — the directory of users, groups, and applications that controls who can authenticate and what they can access. Equivalent: No K8s equivalent — think of it as the IAM backbone of the entire Azure platform

Primary Why & What: Every action in Azure — logging into the portal, calling an API, running a CLI command — is authenticated via Entra ID. It is the source of truth for identities in Azure.

```
Workflow:  Input: Create user/group + assign roles → Process: User
authenticates via Entra ID (MFA, Conditional Access applied) → Output: Access
token issued, user can access only the resources their roles permit
```

Best For: Centralising all employee identities for an organisation — IT creates users in Entra ID, assigns them to groups, and those groups determine their access to Azure resources and Microsoft 365

⚡ **Key Metric: Free tier includes basic features. P1 licence needed for Conditional Access + SSPR. P2 licence for Privileged Identity Management (PIM) and Identity Protection**

Azure Role-Based Access Control (RBAC) [Both]

Concept: A system for granting fine-grained access to Azure resources by assigning roles (Owner, Contributor, Reader, custom roles) to identities (users, groups, service principals) at specific scopes. Equivalent: Kubernetes RBAC — same concept of roles + role bindings

Primary Why & What: Least-privilege access — give users only the permissions they need at the right scope. Without RBAC, everyone would need Owner-level access to do anything.

Workflow: `Input: Select identity + select role + select scope (subscription / resource group / resource) → Process: Azure stores role assignment; every API call is checked against it → Output: Identity can only perform actions the assigned role permits at the defined scope`

Best For: Giving a developer Contributor access to a single resource group (their dev environment) while denying them any access to production resource groups — all within the same subscription

⚡ **Key Metric: RBAC itself has no cost. Billed only if using advanced features like PIM (Entra ID P2 licence)**

Microsoft Entra Domain Services [AZ-900]

Concept: A managed service providing traditional Active Directory features (domain join, Group Policy, LDAP, Kerberos/NTLM auth) without the need to manage domain controller VMs. Equivalent: No K8s equivalent — it is managed Windows Active Directory in the cloud

Primary Why & What: Legacy applications that rely on domain join, LDAP, or Kerberos cannot use modern Entra ID directly. Entra Domain Services bridges on-premises AD capabilities into Azure as a managed service.

Workflow: `Input: Enable Entra Domain Services for a managed domain name + link to VNet → Process: Azure provisions two domain controller VMs (managed, invisible) + syncs identities from Entra ID → Output: VMs in the VNet can join the domain; apps can use LDAP/Kerberos`

Best For: A company migrating an old Java EE application that hard-codes LDAP queries to Active Directory — Entra Domain Services provides an LDAP endpoint in Azure without rebuilding the app

⚡ **Key Metric: Billed per SKU tier (Standard / Enterprise / Premium) per hour — based on object count (users/groups managed)**

◆ GOVERNANCE AND MANAGEMENT TOOLS

Azure Resource Manager (ARM) & ARM Templates [Both]

Concept: ARM is the control plane for all Azure resources — every operation (create, read, update, delete) goes through ARM. ARM Templates are JSON files that declaratively define and deploy Azure infrastructure. Equivalent: Kubernetes control plane + YAML manifests combined

Primary Why & What: Deploy complex multi-resource environments consistently and repeatably from a single JSON file. Eliminates manual portal clicking — infrastructure becomes version-controlled code.

Workflow: `Input: JSON template with resources section + parameters → Process: ARM validates template, resolves dependencies, provisions resources in correct order → Output: All defined resources deployed in the specified state`

Best For: Spinning up an identical copy of the production environment (VNet + VMs + NSGs + Storage) for load testing in minutes — just run the same ARM template with different parameter values

⚡ **Key Metric: ARM Templates are free. Billed only for the resources they deploy**

Azure Bicep [Both]

Concept: A domain-specific language (DSL) that simplifies writing ARM Templates — cleaner syntax, no JSON boilerplate, compiles to ARM JSON. Equivalent: Helm Charts — a higher-level abstraction over raw K8s YAML

Primary Why & What: ARM JSON is verbose and error-prone. Bicep achieves the same result with 40-60% less code, type checking, intellisense in VS Code, and native module support.

Workflow: `Input: Write .bicep file → Process: Bicep CLI compiles to ARM JSON → Deploy via az deployment group create → Output: Same ARM-based resource deployment, but authored in cleaner syntax`

Best For: A DevOps engineer building a reusable infrastructure library — Bicep modules let them write a VNet module once and call it from any project's main.bicep with different parameters

⚡ **Key Metric: Bicep itself is free. Billed only for resources deployed. Bicep is the recommended replacement for raw ARM JSON going forward**

Azure Policy [Both]

Concept: A governance service that enforces organisational rules by evaluating Azure resource properties against policy definitions — can audit, deny, or auto-remediate non-compliant resources. Equivalent: OPA/Gatekeeper in Kubernetes — policy enforcement at admission time

Primary Why & What: Prevent non-compliant resources from being created at scale. Instead of trusting individuals to follow naming conventions or location restrictions, Azure Policy enforces them automatically for every resource.

Workflow: `Input: Define policy (allowed VM SKUs, required tags, allowed regions) + assign to scope → Process: Azure evaluates every resource create/update against assigned policies → Output: Non-compliant resources are denied at creation, or flagged in compliance reports`

Best For: Ensuring all storage accounts in a subscription have HTTPS-only access enabled — Azure Policy denies creation of any storage account without this setting, without any manual review

⚡ **Key Metric: Free for built-in policies. Azure Policy regulatory compliance (guest configuration) billed per server per month**

Azure Key Vault [Both]

Concept: A cloud service for securely storing and managing secrets (passwords, connection strings), cryptographic keys, and certificates. Equivalent: Kubernetes Secrets — but hardened, audited, and integrated with HSM-backed key storage

Primary Why & What: Never store secrets in code, config files, or environment variables. Key Vault centralises secret management with access audit logs, automatic rotation, and hardware security module (HSM) backing.

Workflow: `Input: Store secret/key/certificate in vault + assign access policies (RBAC or vault access policy) → Process: Application authenticates to`

```
Entra ID → receives token → calls Key Vault API → Output: Secret value returned
only to authorised callers; all access logged
```

Best For: An application that needs a database connection string — instead of the string being in appsettings.json in a Git repo, the app fetches it from Key Vault at runtime using its Managed Identity

⚡ **Key Metric: Billed per 10,000 operations + per certificate renewal + per managed HSM hour**

Azure Monitor [Both]

Concept: A comprehensive monitoring platform that collects metrics, logs, and traces from all Azure resources and surfaces them via dashboards, alerts, and workbooks. Equivalent: Prometheus + Grafana + Alertmanager — but managed by Azure

Primary Why & What: Observability is non-negotiable in production. Azure Monitor aggregates telemetry from every Azure service (VM CPU, App Service latency, SQL DTU) and your custom applications into one pane of glass.

```
Workflow:  Input: Resources automatically emit metrics → Configure diagnostic
settings for logs → Define alert rules + action groups → Process: Monitor
evaluates alert conditions on incoming data → Output: Alerts fired to
email/SMS/webhook; dashboards visualise all metrics
```

Best For: Detecting that a VM's CPU has been above 90% for 5 minutes and automatically sending a Teams notification and triggering an Azure Automation runbook to scale out

⚡ **Key Metric: Basic platform metrics: free. Log ingestion billed per GB to Log Analytics. Alert rules billed per rule per month**

Log Analytics Workspace [Both]

Concept: The central data store and query engine within Azure Monitor — ingests logs from VMs, containers, and Azure services, and lets you query them with KQL (Kusto Query Language). Equivalent: Elasticsearch/Loki — a centralised log aggregation and query backend

Primary Why & What: Raw logs from hundreds of VMs and services need to land somewhere queryable. Log Analytics Workspace is that place — structured ingestion with a powerful query language (KQL) for analysis and alerting.

```
Workflow:  Input: Connect VMs via Azure Monitor Agent + enable diagnostic
settings for Azure services → Process: Logs ingested into workspace → Run KQL
queries or set alert rules → Output: Query results, dashboards, and triggered
alerts
```

Best For: Debugging a flapping microservice by running a KQL query across logs from 20 VMs simultaneously to find the exact error message occurring before each crash

⚡ **Key Metric: Billed per GB of data ingested + per GB retained beyond 31 days. Pay-as-you-go or commitment tier (discounted for high volumes)**

◆ SECURITY SERVICES [AZ-900 Coverage]

Microsoft Defender for Cloud [AZ-900]

Concept: A unified cloud security posture management (CSPM) and workload protection platform for Azure, multi-cloud, and on-premises resources. Equivalent: No K8s equivalent — think of it as a security auditor + threat protector combined

Primary Why & What: Continuously assess the security configuration of all your Azure resources, get a Secure Score, and receive actionable recommendations to harden your environment against attacks.

```
Workflow:  Input: Enable Defender for Cloud on subscription → Process: Continuously scans resource configurations, ingests security signals, runs threat detection ML models → Output: Secure Score, security recommendations ranked by impact, real-time alerts for active threats
```

Best For: A company that has just migrated 200 resources to Azure and wants a prioritised list of security misconfigurations to fix first — Defender shows that 6 critical issues (public storage accounts, weak NSG rules) account for 40% of their risk

⚡ **Key Metric: Free tier: basic CSPM and recommendations. Enhanced protections (Defender for Servers, DBs, Containers) billed per resource per month**

Microsoft Sentinel [AZ-900]

Concept: A cloud-native SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation and Response) service. Equivalent: No K8s equivalent — think Splunk / Elastic SIEM as a managed Azure service

Primary Why & What: Collect and correlate security signals from across your entire estate (Azure, Microsoft 365, on-premises, third-party SaaS) to detect, investigate, and respond to threats at scale using AI.

```
Workflow:  Input: Connect data sources via connectors (Azure AD, Office 365, firewalls, etc.) → Data lands in Log Analytics Workspace → Process: Sentinel runs analytics rules + ML models to detect threats → Output: Incidents created for investigation; automated playbooks can auto-respond
```

Best For: Detecting that a compromised user account is logging in from two countries 30 minutes apart (impossible travel) — Sentinel's ML correlates Entra ID sign-in logs and raises a high-severity incident automatically

⚡ **Key Metric: Billed per GB of data ingested into the workspace designated for Sentinel — same Log Analytics pricing, plus Sentinel capacity reservations available**

◆ BACKUP AND DISASTER RECOVERY [AZ-104 Coverage]

Azure Backup (Recovery Services Vault) [AZ-104]

Concept: A centralised, cloud-based backup solution for Azure VMs, on-premises servers (via MARS agent), Azure File Shares, and Azure SQL. Equivalent: No K8s equivalent — think Velero (K8s backup tool) but for Azure workloads

Primary Why & What: Protect production workloads from accidental deletion, ransomware, and hardware failure with policy-driven, encrypted backups stored in Azure — no infrastructure to manage.

```
Workflow:  Input: Create Recovery Services Vault + define backup policy (frequency, retention) + enable backup on VM/File Share → Process: Azure takes scheduled snapshots, stores to vault → Output: Restore points available for VM restore or file-level restore
```

Best For: A production Windows Server VM that must be recoverable within 4 hours — daily backups with 30-day retention, tested monthly via File Restore to verify integrity

⚡ **Key Metric: Billed per protected instance per month + per GB of backup storage consumed in the vault**

Azure Site Recovery (ASR) [AZ-104]

Concept: A Disaster Recovery as a Service (DRaaS) solution that continuously replicates Azure VMs (or on-premises VMs) to a secondary Azure region, enabling failover within minutes. Equivalent: No K8s equivalent — think of it as a real-time VM cloning service across regions

Primary Why & What: Achieve region-level DR with near-zero RPO (Recovery Point Objective). If East US goes down, ASR can fail over your entire application to West US in minutes, with the VM in a consistent last-replicated state.

Workflow: `Input: Enable replication on VM + choose target region + configure recovery plan → Process: ASR continuously replicates VM disk changes to target region in the background → Output: In a DR event, trigger failover — VM boots in target region with replicated data from the last sync point`

Best For: A company whose SLA guarantees 99.99% availability — ASR replicates their critical VMs to a secondary region, and they run quarterly DR drills by doing test failovers without disrupting production

⚡ **Key Metric: Billed per protected VM per month + outbound replication data transfer + storage in target region**

◆ DEVOPS AND ADDITIONAL SERVICES [AZ-900 Coverage]

Azure DevOps [AZ-900]

Concept: An end-to-end DevOps platform offering tools for planning (Boards), source code management (Repos), CI (Build Pipelines), and CD (Release Pipelines). Equivalent: GitLab CI/CD + Jira combined — but Microsoft-hosted

Primary Why & What: Unify all stages of the software delivery lifecycle — from backlog management to deployment — in one platform with deep integration with Azure services.

Workflow: `Input: Team creates work items in Boards → Developers commit code to Repos → Build Pipeline triggers on commit: build + test → Release Pipeline deploys to Azure Web App / AKS → Output: Automated, repeatable delivery from code commit to production`

Best For: A team where developers, testers, and ops are all using different tools — Azure DevOps replaces Jira + GitHub + Jenkins with one integrated platform with native Azure deployment targets

⚡ **Key Metric: Free tier: 5 users + 1800 CI/CD minutes/month. Additional users billed per user/month; self-hosted agents are free**

Azure Logic Apps [AZ-900]

Concept: A low-code / no-code cloud platform for building automated workflows that integrate apps, data, and services — triggered by events. Equivalent: No K8s equivalent — think Zapier / n8n but enterprise-grade on Azure

Primary Why & What: Automate business processes and integrate disparate systems without writing code. Hundreds of pre-built connectors for Azure services, Microsoft 365, Salesforce, Slack, and more.

Workflow: `Input: Define trigger (e.g., new blob uploaded to Storage) → Add connector actions (e.g., send email via Outlook, post to Teams) → Process: Azure hosts and runs the workflow serverlessly → Output: Automated multi-step business process executes on every trigger event`

Best For: Automatically sending a manager approval email every time an invoice file is uploaded to a SharePoint folder — no code, just connectors wired together in the visual designer

⚡ **Key Metric: Billed per action execution. Connector type (standard vs enterprise/ISE) affects per-run cost. First 4,000 built-in actions/month are free on Consumption plan**

Azure Machine Learning [AZ-900]

Concept: A cloud platform for building, training, deploying, and managing machine learning models at scale — supporting both code-first (Python/SDK) and automated ML approaches. Equivalent: No K8s equivalent — MLflow + Kubeflow as a managed Azure service

Primary Why & What: Train and deploy ML models without managing GPU clusters, dependency hell, or experiment tracking infrastructure. Azure ML handles versioning, compute provisioning, model registry, and deployment.

Workflow: Input: Dataset + training script + compute cluster config → Process: Azure ML provisions compute, runs training job, tracks metrics/params → Output: Registered model deployed as real-time endpoint or batch pipeline

Best For: A data science team training a customer churn prediction model on 5 years of data — Azure ML handles GPU cluster provisioning, tracks 200 experiment runs, and deploys the best model as a REST API endpoint

⚡ **Key Metric: Billed per underlying compute (VM hours for training clusters) + storage + endpoint hosting hours. AML workspace itself is free**

Azure AI Services (Cognitive Services) [AZ-900]

Concept: A collection of pre-built AI APIs for vision, speech, language, and decision tasks — consumable via REST API without any ML expertise. Equivalent: No K8s equivalent — AI capabilities as microservice APIs

Primary Why & What: Add AI capabilities (face detection, speech-to-text, sentiment analysis, translation) to any application by calling an API — no data science team required. Azure hosts, trains, and scales the models.

Workflow: Input: API call with image/audio/text payload → Process: Azure runs pre-trained model against input → Output: Structured JSON response (e.g., detected objects + confidence scores, translated text)

Best For: A customer support portal that automatically classifies incoming tickets by sentiment (positive/negative/neutral) and routes angry customers to senior agents — using the Text Analytics API with zero ML expertise needed

⚡ **Key Metric: Billed per API call or per hour of usage (varies by service). Free tier available for most services (e.g., 5,000 text records/month free for Language service)**

Azure Cost Management [AZ-900]

Concept: A built-in Azure tool for visualising, analysing, and controlling cloud spending — with budgets, alerts, cost allocation, and recommendations. Equivalent: No K8s equivalent — think of it as an AWS Cost Explorer equivalent

Primary Why & What: Cloud costs grow invisibly without governance. Cost Management gives teams visibility into spend by service/resource/tag and lets them set budget alerts before overspending happens.

Workflow: Input: View cost analysis by subscription/resource group/service → Set budget with threshold (e.g., $500/month) + alert action (email) → Process: Azure monitors actual vs budgeted spend → Output: Proactive alerts before budget is exceeded; advisor recommendations to cut waste

Best For: A startup that accidentally left 10 GPU VMs running over a weekend — Cost Management's budget alert triggers an email on Friday when spend hits 80% of the monthly budget, prompting the team to check and shut them down

Key Metric: Cost Management tool itself is free. Azure Advisor recommendations are also free

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